

YITIAN WANG

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SUMMARY

Ph.D. scholar in Electrical Engineering specializing in thermal transport and lattice dynamics in functional oxides and solid-state ionic conductors. Experienced in crystal and thin-film growth, vacuum deposition, material characterization, and data-driven modeling using Python and MatLab. Published 5 first-author works in leading journals and skilled at collaborating with national labs and cross-disciplinary device and materials teams. Extensive teaching background, including tutoring since undergraduate years and teaching assistant experience during Ph.D. studies.

EDUCATION

University of California, Riverside, CA Ph.D. in Electrical and Computer Engineering	<i>Mar. 2021 - Jun. 2026 (expected)</i>
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Columbia University in the City of New York, NY Master of Science, Material Science and Engineering	<i>Aug. 2019 - Feb. 2021</i>
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University of California, Berkeley, CA Summer Session, Physics	<i>Jul. 2017 - Aug. 2017</i>
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Beijing Normal University, CN Bachelor of Science, Physics	<i>Sep. 2015 - Jun. 2019</i>
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AWARDS

- Dissertation Completion Fellowship Award (2025), UCR
- Materials Research Society 2023 Spring Highly Commended Student Talk Award
- 'Jingshi' Scholarship (2016 - 2018), Beijing Normal University

SKILLS

Software / Programming

- Python, MATLAB, R, SQL, LaTeX
- COMSOL, SolidWorks, Linux
- GSAS-II, Mantid, ImageJ, VESTA
- Quantum ESPRESSO, VASP
- AI Agent Training and Deployment

Materials Synthesis & Growth

- Floating-Zone Crystal Growth
- Physical Vapor Deposition / Magnetron Sputtering, Chemical Vapor Deposition

Characterization & Spectroscopy

- Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), X-ray Diffraction (XRD), Atomic Force Microscopy (AFM)
- Inelastic Neutron Scattering, Raman, Fourier Transform Infrared Spectroscopy (FTIR)
- Physical Property Measurement System (PPMS), Differential Scanning Calorimetry (DSC), Thermogravimetric Analysis (TGA)
- Four-Probe Transport Methods, Mechanical Strength Testing

Battery & Electrochemistry

- Prototype Cell Assembly, Composite Separator Fabrication
- Electrochemical Impedance Spectroscopy (EIS), Galvanostatic Cycling
- Temperature-dependent Transport Testing, Thermal Abuse Testing

Methods

- Design of Experiments (DOE)
- Uniaxial Strain Tuning

RESEARCH EXPERIENCE

Graduate Research Assistant

University of California

Jun. 2021 - Jun. 2025

Riverside, CA

- Studied thermal transport and ionic mobility in solid-state electrolytes and complex oxides.
- Grew single-crystal oxide Li-ion conductors using a floating-zone furnace.
- Characterized structure and transport with XRD, transport measurements, and electrochemical testing.
- Built a two-channel thermal-conductivity fitting model in MatLab.

Experiment Lead

Oak Ridge National Laboratory

Jan. 2022 - Jul. 2023

Oak Ridge, TN

- Led inelastic neutron-scattering studies on LLZTO using TAX at HFIR and ARCS at SNS.
- Measured phonon spectra of single-crystal solid electrolytes.
- Processed scattering datasets and automated analysis workflows in Mantid and Python.

Student Researcher

Columbia University

Sep. 2019 - Jan. 2021

New York, NY

- Developed composite lithium-ion battery separators with γ - C_3N_4 and PVDF using solution processing.
- Fabricated prototype coin cells and measured impedance and galvanostatic cycling.
- Ran first-principles simulations in Quantum ESPRESSO on defect-engineered olivine structures.

Student Researcher

Beijing Normal University

Sep. 2018 - May 2019

Beijing, CN

- Designed low-temperature uniaxial-strain devices in SolidWorks.
- Characterized detwinning effects in $BaFe_2As_2$ with PPMS.

SELECTED PUBLICATIONS

- Glass-like thermal transport in polycrystalline perovskite lithium-ion conductor $Li_{3/8}Sr_{7/16}Hf_{1/4}Ta_{3/4}O_3$

- *Chemical Communications*, 2025 [\[Link\]](#)
- **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Low Thermal Conductivity and Lattice Anharmonicity of NaSICON-type Solid Electrolyte $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$
 - *Tungsten*, 2025 [\[Link\]](#)
 - **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte: Linking lattice and ionic dynamics with thermal transport
 - *PRX Energy*, 2025 [\[Link\]](#)
 - **Y. Wang**, Y. Su, J. Carrete, H. Zhang, N. Wu, Y. Li, H. Li, J. He, Y. Xu, S. Guo, Q. Cai, D. L. Abernathy, T. Williams, K. V. Kravchyk, M. V. Kovalenko, G. K. H. Madsen, C. Li, X. Chen
- Thermal properties and lattice anharmonicity of Li-ion conducting garnet solid electrolyte $\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$
 - *Journal of Materials Chemistry A*, 2024 [\[Link\]](#)
 - **Y. Wang**, S. Li, N. Wu, Q. Jia, T. Hoke, L. Shi, Y. Li, X. Chen
- Enhanced magnon thermal transport in yttrium-doped spin ladder compounds $\text{Sr}_{14-x}\text{Y}_x\text{Cu}_{24}\text{O}_{41}$
 - *Journal of Applied Physics*, 2024 [\[Link\]](#)
 - S. Li, S. Guo, **Y. Wang**, H. Li, Y. Xu, V. Carta, J. Zhou, X. Chen
- Size-dependent magnon thermal transport in a nanostructured quantum magnet
 - *Cell Reports Physical Science*, 2024 [\[Link\]](#)
 - S. Guo, H. Li, X. Bai, **Y. Wang**, S. Li, R. E. Dunin-Borkowski, J. Zhou, X. Chen
- Crystal structure and thermoelectric properties of layered van der Waals semimetal ZrTiSe_4
 - *Chemistry of Materials*, 2022 [\[Link\]](#)
 - Y. Xu, Z. Barani, P. Xiao, S. Sudhindra, **Y. Wang**, A. A. Rezaie, V. Carta, K. N. Bozhilov, D. Luong, B. P. T. Fokwa, F. Kargar, A. A. Balandin, X. Chen
- Single crystal growth and electrochemical studies of garnet-type fast Li-ion conductors
 - *Tungsten*, 2022 [\[Link\]](#)
 - **Y. Wang**, X. Chen